Source partitions

Shared-exclusions information

Sepehr Mahmoudian

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Development report. The derivations are written by hand and checked by exact rational computation or, where stated, in binary64 arithmetic. Z3 checked the event and linear algebra, not the complete probability theorems. No new Lean proof was accepted for this report. Priority, calibration, deployment benefit and a complete three-source certificate are not claimed.

Revision of 31 October 2026

This revision takes in the results of a dated addendum. The addendum followed an AI-agent review of 28 October 2026 and its adjudication record. No number of the first version changes.

- The identity $D = L + 2R$ holds for every recoding, not only for deterministic ones (Section 3).
- A closed form gives the best lower bound from the seven changes of mutual information, with a proof and 48 exact dual certificates (Section 3). It is larger than the first lower bound on six office distributions.
- The upper bound (1) has a form that uses the target (Section 2). The nonnegative informative and misinformative atoms give an exploratory refinement of the range bound (Section 7).
- Section 7 explains how to read the recorded-data results. The evaluation days differ from the training days, 205 of the 445 score records are computed on training rows, a refitted model changes its smoothing as well as its inputs, and several cells are sparse. Section 7 also reports a third upper bound that the study saved.
- The report uses the standard terms of the pid-rs terminology list. For example, *distribution* replaces the earlier *law*, and *redundancy value* replaces *cumulative*.
- Section 10 corrects the status statement. The report is on the main branch of the repository; the scientific questions that Section 8 lists stay open.

The addendum remains the dated record of these changes and of two repaired replay scripts.

1. Question, definitions and assumptions

Question. A code can change how sources are represented, for example by merging two rare values of a sensor pair, while the information about the target stays almost the same. How much can the partial information decomposition of the sources change then, and which quantities that are cheap to compute give bounds on this change?

Why it matters. Partial information decomposition (PID) splits the information that several sources carry about a target into atoms: information that is redundant, unique to one source or synergistic. A recoding can keep the predictive information and still move information between

atoms. An analysis that attributes information to sources then gives a different answer after the recoding. Bounds on the change tell an analyst, before the atoms are computed, how much a recoding can change the attribution at most, and when it must change it.

Object. The decomposition is the categorical shared-exclusions PID of Makkeh, Gutknecht and Wibral (MGW), arXiv:2002.03356v5. All logarithms are natural, so every information value is in nats; the defining paper uses bits. The recoding keeps every source index and the complete redundancy lattice of the original sources. Signed atoms can be negative. This is not the continuous shared-exclusions PID of Ehrlich et al., and no result transfers to another redundancy measure.

Assumptions. All variables are categorical with finitely many values. A distribution is a set of nonnegative probabilities that sum to one; zero probabilities are allowed. A calculation for a given distribution needs no Gaussian model, no independent rows, no stationarity and no positive lower bound on the probabilities. A statement about a population from observed rows needs a separate sampling model, which this report does not supply.

Terms

Term	Meaning
Source, target	The input variables $S = (S_1, \dots, S_r)$ and the variable T that they inform about.
Recoding	A new representation $S' = (S'_1, \dots, S'_r)$ of the sources with the same source indices. A deterministic recoding is a function $S' = f(S)$.
Coupling	One joint distribution of (S, S', T) whose marginals are the given distributions of (S, T) and (S', T).
Collection, antichain	A collection is a nonempty set of source indices. An antichain is a nonempty family of collections, none of which contains another; its members are its branches.
Redundancy lattice	The antichains, ordered as by Williams and Beer (2010): $\beta \preceq \alpha$ when every branch of α contains a branch of β. Three sources give 18 nodes.
Redundancy value	The MGW shared-exclusions redundancy at a node, in nats: the informative part minus the misinformative part. It can be negative.
Atom	Möbius inversion on the lattice turns the redundancy values into atoms. Each redundancy value is the sum of the atoms at and below its node.
Informative and misinformative atoms	The atoms of the informative part and of the misinformative part. MGW Theorem IV.3 proves that they are never negative. A signed atom is their difference.
Total change of the atoms	$D = \sum_\alpha \lvert\Delta_\alpha\rvert$, where Δ_α is the old signed atom minus the new one at node α.
Variation of information	$V_B = H(S'_B \mid S_B) + H(S_B \mid S'_B)$, a distance between two partitions of the outcomes (Section 2).
Tail mass	The fraction of rows whose recoded pair is not one of the two pairs that the code keeps.

Symbols

Symbol	Meaning and domain
$r, [r], i$	Number of sources ($r \geq 1$), the index set $\{1, \dots, r\}$, one index
S, S', T	Old and new source vectors and the common finite target
P	A coupling of (S, S', T)
B, S_B	A collection $B \subseteq [r]$ and the subvector $(S_i)_{i \in B}$
$\alpha, \beta, \mathcal{L}$	Antichains (nodes) and the redundancy lattice of the r sources
$E^S_\beta(s)$	Source event of node β at the reference outcome s
$G_\beta, C_\beta, \Pi_\alpha$	Mean event surprise, redundancy value and signed atom, in nats
$U_{\alpha\beta}$	Entry of the Möbius matrix: row of atom α, column of redundancy value β
H, I	Shannon entropy and (conditional) mutual information, in nats
V_B, w_B	Variation of information of collection B and its integer weight
Δ_α, D	Old minus new signed atom, and the total change of the atoms, in nats
L_B, L	Change of the mutual information of S_B, and of the whole vector S, with T

Information facts that the proofs use

Let A and B be any finite variables; in this paragraph B is not a collection. The entropy is $H(A) = -\sum_{a:P(a)>0} P(a) \ln P(a)$. The conditional entropy $H(A \mid B)$ is the average, over the values b with positive probability, of the entropy of $P(A \mid B = b)$. The mutual information is $I(A;T) = H(T) - H(T \mid A)$. Outcomes of zero probability are left out of every sum, which is the convention $0 \ln 0 = 0$. Every logarithm that is evaluated has a positive argument.

The proofs use two facts. The chain rule $H(A,B) = H(B) + H(A \mid B)$ follows by expanding conditional probabilities. The inequality $H(A \mid B,T) \le H(A \mid B)$ holds because the difference is a conditional mutual information, an average of sums $\sum p \ln(p/q)$ in which q is the product of the conditional marginals. For $x > 0$, $-\ln x \ge 1 - x$, so $\sum p \ln(p/q) \ge \sum_{p>0}(p - q) \ge 0$. Neither fact assumes that the sources are independent.

2. Main upper bound

The coupling must have *both* given marginals, $P(S,T)$ and $P(S',T)$. A matching of the two source histograms that ignores the target is not enough. For each collection B, the variation of information

$$V_B = H(S'_B \mid S_B) + H(S_B \mid S'_B)$$

measures how much the partition of the outcomes by the values of S'_B differs from the partition by the values of S_B under the coupling; see Meilă (2005). It does not use the target, and it is not a distance between the marginal histograms.

Let U be the Möbius matrix of the lattice (Proof 5 below). Set

$$c_\beta = \sum_\alpha |U_{\alpha\beta}|, \qquad w_B = \sum_{\beta : B \in \beta} c_\beta, \qquad D = \sum_\alpha |\Pi_\alpha(S;T) - \Pi_\alpha(S';T)|.$$

Here $B \in \beta$ means that B is a branch of β. Then

$$\boxed{D \le \sum_{\emptyset \ne B \subseteq [r]} w_B V_B.} \tag{1}$$

For three sources the exact coefficients give

$$\boxed{D \le 22 \sum_{|B| \in \{1,2\}} V_B + V_{\{1,2,3\}} \le 67 \sum_{i=1}^{3} V_{\{i\}}.} \tag{2}$$

The bound needs no positivity of any atoms, no factor for the number of target values and no lower bound on the cell probabilities. Its coefficients grow with the size of the lattice, and the entropy terms can grow with the number of source values. It depends on the number of sources and is not optimal.

Proof 1: the event functional

For a source value s with $P(s) > 0$, define the source event and the mean event surprise

$$E^S_\beta(s) = \bigcup_{B \in \beta} \{u : u_i = s_i \text{ for all } i \in B\}, \qquad G_\beta(S) = \sum_{s:P(s)>0} P(s)[-\ln P(E^S_\beta(s))].$$

An outcome u is in the event when it agrees with s on every index of at least one branch. The event contains the reference outcome s, so its probability is positive; so is its conditional probability given $T = t$ when $P(s, t) > 0$. Define $G_\beta(S \mid T)$ by computing the same functional in each target stratum $T = t$ with $P(T = t) > 0$ and averaging with weights $P(T = t)$. Expanding the MGW logarithmic ratio gives

$$C_\beta(S; T) = G_\beta(S) - G_\beta(S \mid T). \tag{3}$$

The first term is the informative part and the second the misinformative part. The first sums over the target values; the second keeps the weights of the joint distribution of sources and target.

Proof 2: nested unions

Take m pairs of events $C_j \subseteq D_j$ with $P(C_j) > 0$, where $j = 1, \dots, m$ labels branches. Write $c_j = P(C_j)$, $d_j = P(D_j)$, $c = P(\bigcup_j C_j)$ and $d = P(\bigcup_j D_j)$. The set difference of the two unions lies in the union of the differences $D_j \setminus C_j$, so the union bound gives

$$d - c \le \sum_j (d_j - c_j).$$

Because $c \ge c_j > 0$,

$$\frac{d}{c} \le 1 + \sum_j \frac{d_j - c_j}{c} \le 1 + \sum_j \left(\frac{d_j}{c_j} - 1 \right) \le \prod_j \frac{d_j}{c_j}. \tag{4}$$

For the last step expand the product of the factors $1 + a_j$, where $a_j = d_j/c_j - 1 \ge 0$; the cross terms that the sum leaves out are nonnegative. Thus $0 \le \ln(d/c) \le \sum_j \ln(d_j/c_j)$. The branches may overlap or depend on each other.

Proof 3: deterministic refinement

Let a fine vector F be coarsened coordinate by coordinate to A, with the same source indices: each A_i is a function of F_i. At a value f with $P(f) > 0$, write a for its coarse value. For a branch B, put $C_B = \{F_B = f_B\} \subseteq D_B = \{A_B = a_B\}$. Apply (4), take logarithms and average over f:

$$0 \le G_\beta(F) - G_\beta(A) \le \sum_{B \in \beta} H(F_B \mid A_B). \tag{5}$$

Indeed, the mean logarithmic ratio of one branch is

$$\sum_f P(f) \ln \frac{P(A_B = a_B)}{P(F_B = f_B)} = \sum_{f_B} P(f_B)[-\ln P(F_B = f_B \mid A_B = a_B)].$$

The coordinates outside B sum out, and because A_B is a function of F_B, the ratio inside the logarithm is a conditional probability. The right side is the conditional entropy $H(F_B \mid A_B)$.

Repeat the argument in each target stratum $T = t$ with positive probability and average over t. The increase of the misinformative part lies between zero and $\sum_{B \in \beta} H(F_B \mid A_B, T)$, which is no larger than the bound in (5). No ratio is formed in a stratum of zero probability.

Proof 4: a common refinement

Put $Z_i = (S_i, S'_i)$ at each source index i. The paired vector Z still has r sources, and it coarsens coordinate by coordinate both to S and to S'. Set

$$a_\beta = \sum_{B\in\beta} H(S'_B \mid S_B), \qquad b_\beta = \sum_{B\in\beta} H(S_B \mid S'_B).$$

By (5) with $F = Z$ and $A = S$, the increase $G_\beta(Z) - G_\beta(S)$ lies in $[0, a_\beta]$, because $H(Z_B \mid S_B) = H(S'_B \mid S_B)$; so does the increase of the misinformative part. By (3), the change $C_\beta(Z;T) - C_\beta(S;T)$ is the difference of these two numbers. Two numbers in the same interval differ by at most its length, so the absolute value of the change is at most a_β; there is no extra factor two. The same argument with S' gives b_β, so

$$|C_\beta(S;T) - C_\beta(S';T)| \le a_\beta + b_\beta = \sum_{B\in\beta} V_B. \tag{6}$$

Only conditional Shannon entropies were used; no monotonicity of the signed decomposition was assumed.

A form that uses the target. The proof gives slightly more. Write $a_{\beta,T} = \sum_{B\in\beta} H(S'_B \mid S_B, T)$ and $b_{\beta,T} = \sum_{B\in\beta} H(S_B \mid S'_B, T)$. From S to Z the informative part rises by some $u \in [0, a_\beta]$ and the misinformative part by some $u_T \in [0, a_{\beta,T}]$. From S' to Z they rise by some $v \in [0, b_\beta]$ and $v_T \in [0, b_{\beta,T}]$. The difference $C_\beta(S;T) - C_\beta(S';T)$ equals $(v - v_T) - (u - u_T)$, so it lies in the interval $[-a_\beta - b_{\beta,T},\ b_\beta + a_{\beta,T}]$, and

$$D \le \sum_\beta c_\beta \max\{a_\beta + b_{\beta,T},\ b_\beta + a_{\beta,T}\}. \tag{12}$$

Conditioning cannot increase entropy, so $a_{\beta,T} \le a_\beta$ and $b_{\beta,T} \le b_\beta$: bound (12) is never weaker than bound (1). It needs no deterministic recoding.

Proof 5: Möbius inversion on the unchanged lattice

Order the nodes so that lower nodes come first. Let M be the zeta matrix, with $M_{\alpha\beta} = 1$ when $\beta \preceq \alpha$ and 0 otherwise. It is lower triangular with unit diagonal, and $C = M\Pi$ because each redundancy value is the sum of the atoms at and below its node. In the following recurrence i, j, k index ordered nodes, not sources, and $\mathbf{1}_{i=j}$ is one if $i = j$ and zero otherwise. The inverse $U = M^{-1}$, the Möbius matrix, satisfies

$$U_{ij} = \mathbf{1}_{i=j} - \sum_{k<i} M_{ik} U_{kj}. \tag{7}$$

A finite average commutes with this fixed linear map, so $\Pi = UC$ also for averaged values. Define $\Delta C_\beta = C_\beta(S;T) - C_\beta(S';T)$. The triangle inequality and (6) give

$$D \le \sum_{\alpha,\beta} |U_{\alpha\beta}|\,|\Delta C_\beta| \le \sum_\beta c_\beta \sum_{B\in\beta} V_B = \sum_B w_B V_B.$$

This proves (1); with the interval of the target form in place of (6), the same steps prove (12).

The three-source integer matrix was checked in both directions, $MU = UM = I$. Its column weights are:

Node family	Number of nodes	c_β each
Three singleton branches	1	8
Two singleton branches	3	4
A singleton and the complementary pair	3	4
Three pair branches	1	8
Two pair branches	3	4
One singleton	3	2
One pair	3	2
The full collection	1	1

The weights sum to 65. A fixed singleton is a branch of nodes whose weights sum to $8+4+4+4+2 = 22$; the same holds for a fixed pair. The full collection is a branch only of the top node. By the chain rule, $V_B \le \sum_{i\in B} V_{\{i\}}$. Each source belongs to one singleton, two pairs and the full collection, so its total weight is $22 + 22 + 22 + 1 = 67$; this proves (2).

3. Lower bounds from mutual information

For each collection B, define the changes

$$L_B = I(S_B;T) - I(S'_B;T), \qquad L = L_{[r]}, \qquad \Delta_\alpha = \Pi_\alpha(S;T) - \Pi_\alpha(S';T).$$

Let $J_B = \{\alpha \in \mathcal{L} : \alpha \preceq \{B\}\}$ be the set of nodes at or below the node $\{B\}$, which has the single branch B. Its source event is $\{S_B = s_B\}$, so its redundancy value is $H(S_B) - H(S_B \mid T) = I(S_B;T)$. Each redundancy value is the sum of the atoms at and below its node, and every node lies at or below the top node $\{[r]\}$. Therefore

$$L_B = \sum_{\alpha\in J_B} \Delta_\alpha, \qquad L - L_B = \sum_{\alpha\notin J_B} \Delta_\alpha.$$

The triangle inequality, applied to the two disjoint sums, proves

$$\boxed{\max_{\emptyset\neq B\subseteq[r]} \left(|L_B| + |L - L_B|\right) \le D.} \tag{8}$$

For three sources the seven collections give seven changes, from fourteen values of mutual information. A lower bound above a declared tolerance rejects a claim that the recoding preserves the atoms. A small lower bound cannot confirm such a claim. For a general coupling, L can have either sign.

Lost information and moved information. Let $R = \sum_\alpha \max(-\Delta_\alpha, 0)$ be the total of the negative changes. The sum of all atom changes is the change of the top redundancy value, $I(S;T) - I(S';T) = L$. Splitting the changes into positive and negative parts gives

$$\sum_\alpha \Delta_\alpha = L, \qquad D = L + 2R. \tag{9}$$

This identity holds for every pair of distributions on the same lattice. The first version stated it only for a deterministic recoding; determinism adds only $L \ge 0$. Indeed, when $S' = f(S)$, the chain rule gives $I(S;T) = I(S,S';T) = I(S';T) + I(S;T \mid S')$, so $L = I(S;T \mid S') \ge 0$. The total change of the atoms therefore includes information that moves between atoms, not only information that is lost.

The best lower bound from the seven changes

Bound (8) splits the atoms into two groups. All seven changes together give a larger bound, and its exact value has a closed form. Put $L_\emptyset = 0$ and collect the seven equations $\sum_{\alpha \in J_B} \Delta_\alpha = L_B$ as $A\Delta = L$, where A is the 7×18 matrix with $A_{B\alpha} = 1$ when $\alpha \in J_B$ and 0 otherwise. A maximal chain of collections is a sequence $\{i\} \subset \{i,j\} \subset \{1,2,3\}$ with $i \neq j$; there are six.

Proposition. For every vector $L \in \mathbb{R}^7$ of changes,

$$\min_{d \in \mathbb{R}^{18}:\, Ad = L} \sum_\alpha |d_\alpha| = \max_{i \neq j} \bigl(|L_{\{i\}}| + |L_{\{i,j\}} - L_{\{i\}}| + |L_{\{1,2,3\}} - L_{\{i,j\}}| \bigr). \tag{13}$$

The true change vector satisfies $A\Delta = L$, so the right side is a lower bound on D.

Proof that the minimum is at least the maximum. Along a maximal chain the node sets are nested, $J_{\{i\}} \subseteq J_{\{i,j\}} \subseteq J_{\{1,2,3\}}$, and the last one is the whole lattice. So $L_{\{i\}}$, $L_{\{i,j\}} - L_{\{i\}}$ and $L_{\{1,2,3\}} - L_{\{i,j\}}$ are sums of d over three disjoint node sets. The triangle inequality bounds each absolute value by the sum of $|d_\alpha|$ over its set, and the three sets are disjoint.

Proof that the minimum is at most the maximum. Define a number $v(B)$ for every set of source indices, by increasing size: $v(\emptyset) = 0$ and

$$v(B) = \max_{C \subsetneq B} \bigl(v(C) + \max\{L_C - L_B, 0\} \bigr).$$

Put $u(B) = L_B + v(B)$.

1. *Both functions are nondecreasing.* For $C \subsetneq B$ the definition gives $v(B) \geq v(C)$. It also gives $v(B) \geq v(C) + L_C - L_B$, so $u(B) = L_B + v(B) \geq L_C + v(C) = u(C)$. Both vanish at $\emptyset$, so both are nonnegative.
2. *A nondecreasing set function is a sum of nonnegative atoms.* Let $f \geq 0$ be nondecreasing with $f(\emptyset) = 0$. For a level $x \geq 0$, the sets B with $f(B) > x$ form a family that is closed upwards. Its minimal members form an antichain α_x, which is a lattice node when the family is not empty. A set B belongs to the family exactly when a member of α_x is contained in B, that is, exactly when $\alpha_x \in J_B$. Give each node α the weight $p_\alpha \geq 0$ equal to the length of the set of levels x with $\alpha_x = \alpha$. Then $\sum_{\alpha \in J_B} p_\alpha$ is the length of the levels below $f(B)$, which is $f(B)$, and $\sum_\alpha p_\alpha = f(\{1,2,3\})$.
3. *A feasible vector of the right size.* Apply step 2 to u and to v, and put $d = p(u) - p(v)$. Then $Ad = u - v = L$, and
$$\sum_\alpha |d_\alpha| \leq u(\{1,2,3\}) + v(\{1,2,3\}) = L_{\{1,2,3\}} + 2v(\{1,2,3\}).$$
4. *The value.* Unrolling the definition, $v(\{1,2,3\})$ is the largest total decrease of L along a chain of strictly increasing sets from $\emptyset$ to $\{1,2,3\}$. Inserting an intermediate set never lowers a total decrease, because $\max\{x - z, 0\} \leq \max\{x - y, 0\} + \max\{y - z, 0\}$. So the largest value occurs on a maximal chain. On any chain, the total variation of L equals its net change $L_{\{1,2,3\}}$ plus twice its total decrease. So the bound of step 3 equals the largest total variation over the six maximal chains, which is the right side of (13). This proves the proposition.

The same proof holds for any number of sources, with the maximal chains of all sets of source indices.

Exact dual certificates. For a chain (i,j) and signs $s_1, s_2, s_3 \in \{-1, 1\}$, put

$$y = (s_1 - s_2)\, e_{\{i\}} + (s_2 - s_3)\, e_{\{i,j\}} + s_3\, e_{\{1,2,3\}}.$$

For a node in $J_{\{i\}}$, $(A^\top y)_\alpha = s_1$; for a node in $J_{\{i,j\}}$ but not in $J_{\{i\}}$ it is s_2; otherwise it is s_3. So $|(A^\top y)_\alpha| \le 1$ for every node, and for every vector d with $Ad = L$, $y^\top L = (A^\top y)^\top d \le \sum_\alpha |d_\alpha|$; with $d = \Delta$ this gives $y^\top L \le D$. The program `verification/chain_lower_certificate.py` checks all 48 such vectors exactly on the 18 nodes. It evaluates each vector at the saved outward interval endpoints of the seven changes and keeps the largest result. The bounds are rational numbers, and no floating-point number enters them. They exceed bound (8) on six of the 30 distributions and equal it on the others:

Distribution	Bound (8)	Bound (13)	D
Office, 4 February 2015	0.079376	0.079596	0.765299
Office, 5 February 2015	0.381779	0.394514	0.927480
Office, 6 February 2015	0.068741	0.119510	0.568537
Office, 9 February 2015	0.540707	0.541428	1.549248
Office, 10 February 2015	0.097649	0.150664	0.514408
Office validation	0.402366	0.434641	1.336128

Outward intervals. To bound an absolute value from below, use the distance of its interval from zero. Subtract interval endpoints outward before the second absolute value is taken. Absolute values of midpoints are unsafe. Exact rational products identify a zero; an interval that contains zero gives no permission to set an atom to zero.

4. Exact invariance and a recoverable-tuple counterexample

For a fixed deterministic map f on the source values that occur, the following two statements are equivalent.

1. The map preserves the equality partition of every coordinate: for occurring values s, u, $s_i = u_i$ if and only if $f_i(s) = f_i(u)$.
2. Every original signed atom is unchanged for every finite target kernel. Binary targets already suffice.

For sufficiency, the preserved coordinate matches preserve the conjunctions within branches and the disjunctions across branches. So every event probability, every conditional event probability, the inversion and the averages are unchanged. Statement 1 says that each new coordinate is a one-to-one relabelling of the old values of that coordinate. This is the re-encoding premise of Matthias et al., v2, Property 2.

For necessity, suppose an old coordinate value has two states that the new coordinate separates. Choose the binary target that indicates one of those two new values. The new coordinate determines the target, but the old one leaves positive conditional entropy at the merged value, so the mutual information of that singleton changes. For a violation in the other direction, use an indicator of one old value. A fixed sum of original atoms therefore changes, which contradicts complete invariance. The equivalence is not an invariance test for one actual target, and it is not a converse for random recodings. A constant target makes every signed atom zero.

A counterexample. Let $S_1 = X = 0$, let $Y = (S_2, S_3)$ be 01 with probability δ and 11 with probability $1 - \delta$, where $0 < \delta < 1$, and let $T = S_2$. Keep the pairs 00 and 11, map 01 to 00, and keep X and T. Training counts $00 : 50$, $11 : 49$, $01 : 1$ could select those kept pairs; they are not selected again on evaluation data.

Before the recoding only S_2 varies, and $S_2 = T$. After it, $S_2' = S_3' = T$. The pair stays recoverable on this support: $H(Y \mid Y') = H(Y \mid X, Y') = L = 0$. Before the recoding, the informative redundancy

values are the incidence column of the node $\{\{2\}\}$ in the zeta matrix, times the target entropy; after it they use the node $\{\{2\},\{3\}\}$. Every misinformative value is zero, because T determines every source. Define $h_2(x) = -x\ln x - (1-x)\ln(1-x)$. Inversion gives

$$\Pi_{\{\{2\}\}} = h_2(\delta), \qquad \Pi'_{\{\{2\},\{3\}\}} = h_2(\delta), \qquad D = 2h_2(\delta) > 0.$$

All other signed atoms are zero. Bound (8) for $B = \{3\}$ attains this value, because $L_{\{3\}} = -h_2(\delta)$ and $L - L_{\{3\}} = h_2(\delta)$. Exact checks covered $\delta = 1/2, 1/100, 1/10000$. A recoding that keeps the joint information therefore cannot guarantee that the atoms stay unchanged, even when the tail mass is arbitrarily small.

5. An exact comparison when the recoded pair takes two values

Suppose the recoded pair $Y = (S_2, S_3)$ takes exactly two values with positive probability, and the code $g(Y)$ maps them to two different values, while $X = S_1$ and T are kept. Let $J \subseteq \{2,3\}$ be the set of coordinates in which the two old values differ, and J' the same set for the two new values. Both sets are nonempty. At a reference outcome with positive probability, put $E = \{X = x\}$ and $F = \{Y = y\}$.

For a coordinate j in which the two values differ, the event $\{S_j = s_j\}$ equals F. For a constant coordinate the event is certain. No assumption that all four Boolean match patterns occur is used. So every node event is one of five events: the certain event, $E \cup F$, E, F and $E \cap F$. Their redundancy values are $0, R_2, A, B, C$, with $A = I(X;T)$, $B = I(Y;T)$ and $C = I(X,Y;T)$, where R_2 is the redundancy value of the event $E \cup F$. Set $U_X = A - R_2$, $U_Y = B - R_2$ and $Q = C - A - B + R_2$. These are four intermediate MGW quantities, not a regrouped final estimand.

Define the original nodes

$$R_J = \{\{1\}\} \cup \{\{j\} : j \in J\}, \quad X_0 = \{\{1\}\}, \quad C_J = \{\{j\} : j \in J\}, \quad K_J = \{\{1,j\} : j \in J\}.$$

In the order R_J, X_0, C_J, K_J, the five event forms have the incidence patterns

$$0000, \quad 1000, \quad 1100, \quad 1010, \quad 1111.$$

For a node α whose event has a given form, a digit is 1 when the node at that position lies at or below α. The four digits therefore record, in turn, whether every branch of α contains 1 or meets J; whether every branch contains 1; whether every branch meets J; and whether every branch does both. Placing R_2, U_X, U_Y, Q as atoms at the four nodes R_J, X_0, C_J, K_J reconstructs every original redundancy value $0, R_2, A, B, C$. Inversion is unique because the zeta matrix is unit triangular, so this is the complete atom vector.

A one-to-one recoding of the pair keeps these four quantities. If $J = J'$, all original atoms agree. Otherwise only the node X_0 is common, and the other three values move to disjoint original nodes, which proves

$$\boxed{D = 2(|R_2| + |U_Y| + |Q|), \quad J \neq J'.} \tag{10}$$

If the pair takes only one value, only the atom of the kept source is nonzero and $D = 0$. A map that merges the two values, or a pair with three values, does not satisfy (10). Within target strata that contain one value of the pair, keep the complete lattice and average before taking absolute values. This is a written conditional result; the correspondence between the source paper and a formal statement remains open.

6. Recorded-data design and exact arithmetic

The study uses two public recordings: UCI864 Room Occupancy Estimation (DOI 10.24432/C5P605) and UCI357 Occupancy Detection (DOI 10.24432/C5X01N), both under CC BY 4.0. Their labels were not audited, and their time zone is unknown.

Recording	Original source order	Original target	Training; validation; test
Room	S1_Light, S6_PIR, S7_PIR	Count 0,1,2,3	22–25 Dec 2017; 26 Dec; 10–11 Jan 2018
Office	Light, CO2, Temperature	Occupancy 0,1	2–8 Feb 2015; 9–10 Feb; 11–18 Feb

Binary sources. Sort the n training values of a reading and take the value at one-based rank $\lfloor (n+1)/2 \rfloor$, the lower median. Encode 1 for a value strictly above it and 0 otherwise, including ties. The room light threshold is 1 lux; the passive infrared (PIR) inputs are already bits. The office thresholds are 0 lux, 455 parts per million (ppm) and 20.745 °C. This defines categorical inputs; it is not a PID of the original continuous readings. The run stops on an invalid or missing row, a duplicate timestamp or an unexpected date; it does not drop or repair such rows. The room recording keeps its seven local dates, although the provider's description names four days.

The code. Rank the training values of the pair $Y = (S_2, S_3)$ by count, with lexical order for ties. Keep the two most frequent pairs and map every other pair to the first. Both recordings keep 00 and 11. Each row keeps its X and T, and merged rows add their counts. The joint distribution of X and T is therefore unchanged, and the total variation distance between the old and the new distribution of (S, T) equals the tail mass. The code, the thresholds and the predictors are fitted on training days only and then frozen. There are 24 daily and 6 overlapping pooled distributions: they come from two recordings, not from 30 data sets.

Exact redundancy values. For N rows, let c_z be the count of the joint outcome z, and consider each z with $c_z > 0$ as a reference outcome. Let u_z be the number of rows in its source event, v_z the number of those rows that also have the target value of z, and t_z the number of rows with that target value. All three are positive at a reference outcome. Each redundancy value equals

$$N^{-1} \ln \prod_{z : c_z > 0} \left(\frac{N v_z}{u_z t_z} \right)^{c_z}.$$

The informative and misinformative parts use the factors N/u_z and t_z/v_z. Integer inversion keeps exact rational products, and comparing a product with one decides the sign or a zero. All 108 keys of stage and component hold paired records before and after the recoding, so there are 216 numerical values, not 108 atoms.

Logarithm enclosures. For a rational $x > 0$, write $x = 2^k y$ with an integer k and $1 \le y < 2$, and put $z = (y-1)/(y+1)$. Then $0 \le z < 1/3$, and

$$\ln y = 2 \sum_{i=0}^{63} \frac{z^{2i+1}}{2i+1} + R_{64}, \qquad 0 \le R_{64} \le \frac{2 z^{129}}{129(1-z^2)}.$$

To obtain this remainder, replace the later denominators by 129 and sum the geometric series. The same series at $z = 1/3$ encloses $\ln 2$. A negative k swaps the scaled endpoints. Endpoints are rounded outward onto the denominator 2^{80}, signed coefficients are propagated, and every interval must have width at most 10^{-12} nats. Floating-point values are for display only. The PID distributions are not

smoothed, nothing is set to zero, and no accuracy setting, guard, kernel or default of the library changes.

Separate implementations compare exact keyed products. Prime exponents postpone the arithmetic; ordinary normalized fractions remain. The old and new test cases agree. Two runs that stopped at a 180-second limit remain recorded as failures. Complete runs took 22.29 and 22.19 seconds on an M4 Max with 128 GiB, with peak process memory of 80.4 and 82.9 MB. All 121 output files agree between ordinary and optimized Python, as do 29 distributions from an earlier incomplete run. The times describe these runs; they are not a controlled speed comparison.

Models and actual byte costs

Four lookup models with add-one (Laplace) smoothing are fitted: the full model $F(T \mid X, Y)$, the refitted model on recoded inputs $G(T \mid X, g(Y))$, a model of the kept source alone and a model of the target alone. The full model is also evaluated, unchanged, at recoded inputs. The predicted probability is $(a_t + 1)/(a + m)$, with the training count a_t of class t in the context, the training total a of the context, and the fixed number of original classes $m = 4$ or 2. An unseen context receives the uniform distribution. There is no tuning or model selection.

The scores are the mean negative logarithm of the predicted probability of the observed class and the multiclass Brier score $\sum_t (q_t - \mathbf{1}_{t=T})^2$, with predicted probabilities q_t. They differ from the information loss $L = H(T \mid X, g(Y)) - H(T \mid X, Y)$ of a known distribution. A class that is absent from an evaluation set has no class score.

Three containers encode the binary sources: one kept bit plus a fixed two-bit pair, a lossless dictionary, or a lossy one-bit pair. Each byte count includes the header, codebook, names, thresholds, row count and padding, and decoding is checked. Targets, timestamps, raw readings, shared software and acquisition costs are excluded. The dictionary of the test stream is offline coding of the features with its overhead transmitted; it is not a fitted map.

7. Results and limitations

The numbers below are rounded nats. Exact counts, products and outward intervals remain in the machine-readable records. The upper-bound check was made after the study: all 30 distributions and 2,700 node inequalities were checked, not a selection. In the table, L is the lost mutual information and bound (8) is the lower bound of Section 3.

Distribution	Rows	Tail mass	D	L	Bound (8)
Room training	7,021	585/7021	0.309875	0.097702	0.224412
Room validation	1,063	0	0	0	0
Room test	2,045	92/2045	0.341754	0.044689	0.210829
Office training	8,794	707/4397	0.421590	0.020949	0.136466
Office validation	2,014	549/1007	1.336128	0	0.402366
Office test	9,752	4783/9752	0.582280	0	0.291415

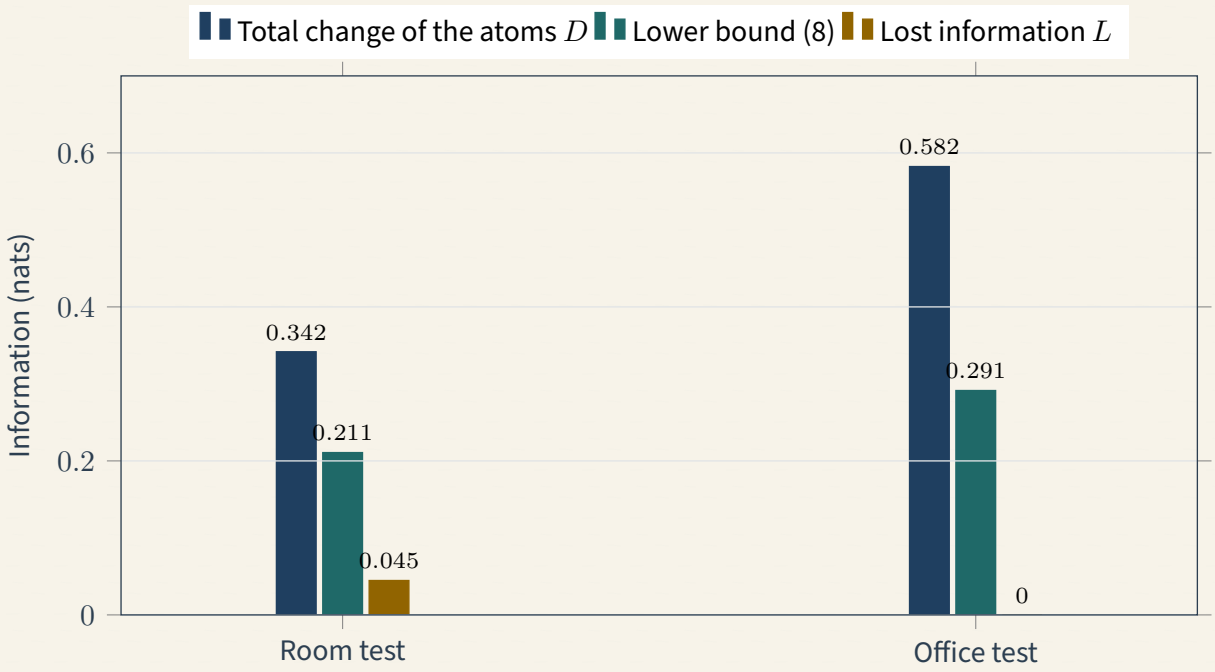


Figure 1: Pooled test distributions: total change of the atoms, the lower bound from mutual information and the lost joint information. All are in nats, but they answer different questions.

A range bound. A fair comparison uses one more published fact, MGW Theorem IV.3: the informative and misinformative atoms are nonnegative, and they sum to $H(S)$ and $H(S \mid T)$, the informative and misinformative values of the top node. Each signed change is $\Delta_\alpha = (\pi^+_\alpha - \pi^-_\alpha) - (\pi'^+_\alpha - \pi'^-_\alpha)$, so the triangle inequality gives

$$D \le H(S) + H(S \mid T) + H(S') + H(S' \mid T) \le 2[H(S) + H(S')]. \tag{11}$$

Recording and role	Upper bound (1)	Range bound (11)
Room training	26.956702	3.408395
Room validation	0	1.911905
Room test	16.650180	2.311007
Office training	49.429024	5.059041
Office validation	43.983542	3.359736
Office test	27.776867	4.354175

On every distribution with $D > 0$, all 21 of them and five of the pooled ones, the range bound is smaller than bound (1). On room validation, where $D = 0$, bound (1) is exactly zero. Bound (1) is not a useful small numerical bound here; its value lies in this exact zero for a one-to-one relabelling and in its sensitivity to partitions. Superiority or optimality of either bound is not shown. The target form (12) improves bound (1) strictly on 13 of the 30 distributions, by small amounts (room test: from 16.650 to 16.582 nats), and it stays above the range bound on every distribution with $D > 0$.

A refinement from nonnegative atoms. For the node $\{B\}$, the informative value is $H(S_B)$ and the misinformative value is $H(S_B \mid T)$. The informative atoms π^+ and the misinformative atoms π^- are nonnegative, so each of the four atom vectors of the old and the new distribution lies in a polytope $\{\pi \in \mathbb{R}^{18} : \pi \ge 0,\ A\pi = h\}$, where h_B is $H(S_B)$, $H(S_B \mid T)$, $H(S'_B)$ or $H(S'_B \mid T)$. Hence D lies between the smallest and the largest value of $\sum_\alpha |\Delta_\alpha|$ over the product of the four polytopes. Only 28 entropies of source sets are needed. The top row of A is all ones, so the largest value is never

above the range bound (11). The smallest value equalled bound (13) on all 30 distributions. The largest value was computed as a mixed-integer linear program with the HiGHS solver in binary64 arithmetic, so this polytope maximum is exploratory, not a certified bound:

Pooled distribution	D	Range bound (11)	Polytope maximum
Room training	0.309875	3.408395	2.267403
Room validation	0	1.911905	0
Room test	0.341754	2.311007	1.613103
Office training	0.421590	5.059041	5.035341
Office validation	1.336128	3.359736	3.281032
Office test	0.582280	4.354175	3.747458

On the 21 distributions with $D > 0$ it is lower than the range bound by 0.5% (office training) to 40% (office, 10 February 2015). The bound stays far above D: the entropies of source sets say nothing about which events the atoms share. The program is kept as `verification/component-sandwich-exploratory.py.txt`; it needs NumPy and SciPy and is not a repository command.

A third saved bound. The study runner saves a third upper bound for each distribution, under the key `continuity_upper_interval`. It computes this bound as $130 \min\{\ln 8,\ \delta \ln(8/\delta) + 8\delta\}$. Here δ is the tail mass, 8 is the number of possible source tuples and 130 is twice the weight sum 65. It comes from an earlier unpublished note on rare-tuple compression, and this report does not derive it; the run only checks that it lies above D. On the 21 distributions with $D > 0$ it lies between 49.3 and 270.3 nats, so it says nothing useful about these distributions.

Prediction scores. The next figure and the table after it give the mean logarithmic loss in nats per row; the Brier scores in the full output have no unit.

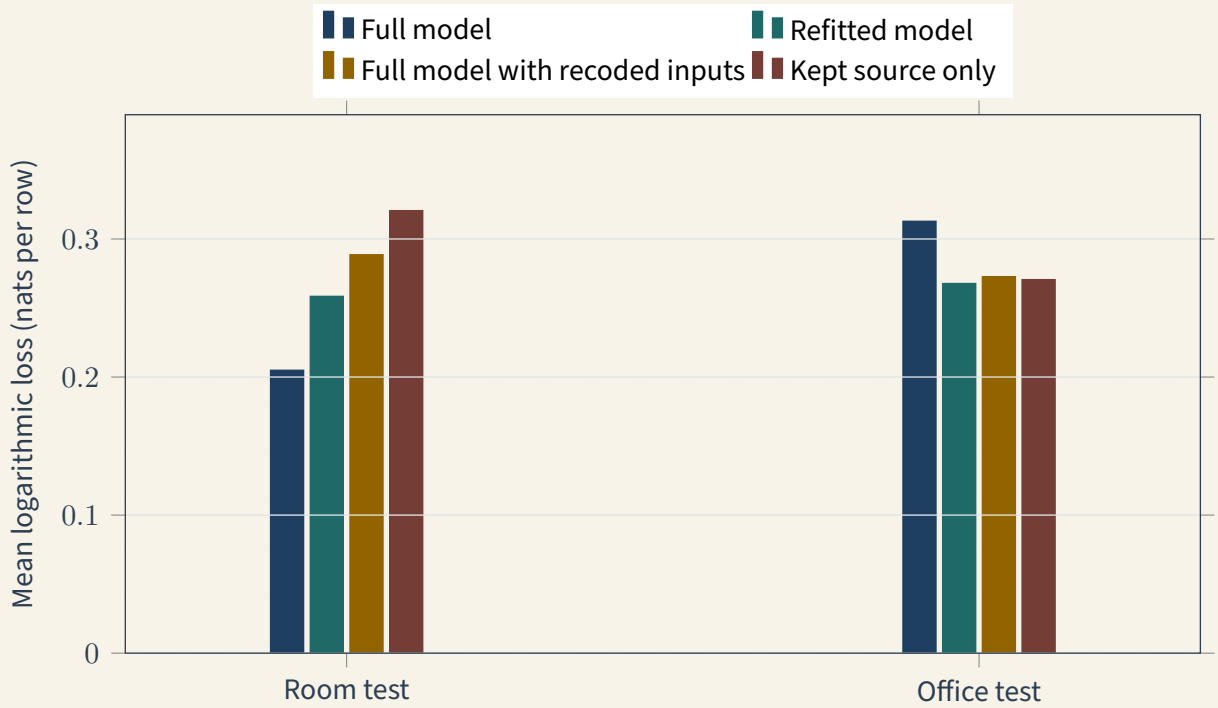


Figure 2: Frozen predictors trade losses differently; lower is better. The target-only model and all class scores remain in the full output.

Model	Room test	Office test
Full model	0.204922	0.312825
Refitted model	0.258447	0.267742
Full model at recoded inputs	0.288580	0.272705
Kept source only	0.320536	0.270571

Room test has no rows of target class 1; all four classes stay in the output. Refitting worsens the room mean test loss and improves the office mean test loss. The office occupied-class loss nevertheless worsens from 0.148113 to 0.188528. Room validation contains only target 0: its empirical signed atoms and D and L are zero, yet the refitted loss worsens from 0.057204 to 0.094076. An invariance statement about the evaluation distribution does not fix a predictor that was fitted on a differently transformed training distribution. There are no population, causal, calibration or statistical-superiority conclusions.

How to read the recorded-data results

The evaluation days differ from the training days. The code and the thresholds were fitted on the training days, and the data moved on later days:

Recording and role	Rows	Tail mass	Constant sources
Room training	7,021	0.083	none
Room validation	1,063	0	S_2, S_3
Room test	2,045	0.045	none
Office training	8,794	0.161	none
Office validation	2,014	0.545	none
Office test	9,752	0.490	S_2

On every office test row the carbon-dioxide bit is 1, because the training median (455 ppm) lies below all test values. About half of the office validation and test rows therefore fall outside the two kept pairs, against 16.1% in training. The office changes measure a code frozen on training data under this shift, not compression in general.

Why no information is lost on the office validation and test days. On the test days only the pairs 10 and 11 occur, and the code maps them to 00 and 11, so it is one-to-one there. On the validation days the code merges 10 with 00, but only rows with the light off ($S_1 = 0$) have both pairs, and all of those rows are unoccupied. Where occupancy varies ($S_1 = 1$), only 10 and 11 occur. In both cases the recoded sources carry the same information about occupancy, so $L = 0$ exactly, while D is large: by (9), the whole change is information that moves between atoms.

Scores on training rows are in-sample. Of the 445 score records, 205 are computed on training rows, 65 on validation rows and 175 on test rows. No record contains a context that is absent from training, so the uniform distribution for unseen contexts is never used.

Refitting also changes the smoothing. Each lookup model adds one pseudocount per class to each context. Merging contexts before fitting gives the merged context one pseudocount per class, where the original contexts had one each. A comparison of the full model with the refitted model therefore changes the inputs, the counts and the shrinkage together.

Why refitting worsens the room validation score. All room validation rows are unoccupied and have the pair 00; 196 of the 1,063 rows have the light on. In training, the rows with the light on and a merged pair are 570 occupied out of 581, while the rows with the light on and the pair 00 are 668 occupied out of 2,538. Refitting merges the two groups, so the predicted probability of an empty room for the light on and the pair 00 falls from 0.7360 to 0.6026. The loss on each of those 196 rows rises from 0.3065 to 0.5065, and $(196/1063) \times 0.2000 = 0.0369$. For the 867 rows with the

light off the predicted probability changes by less than 10^{-5}. This accounts for the observed change from 0.0572 to 0.0941 to four decimal places.

Sparse cells. The room training distribution occupies 21 of its 32 cells, and six cells hold at most five rows. The room test distribution occupies 12 cells, two of them with a single row, and the office test distribution occupies 7 of 16 cells, one with a single row. Atoms computed from such counts are exact for these finite distributions, but as estimates of a population they are biased. For two independent fair bits observed twice, for example, the expected empirical mutual information is $\frac{1}{4}\ln 2 > 0$, although the true value is zero.

A recorded example with two values of the pair

Office test has the pair 10 on 4,783 rows and the pair 11 on 4,969 rows. The carbon-dioxide bit is always 1; the physical concentration need not be constant. The map $10 \mapsto 00$, $11 \mapsto 11$ makes the new bit at the carbon-dioxide index copy the temperature bit. The map is one-to-one on this observed support: the joint predictive information is unchanged, but the attribution to sources moves.

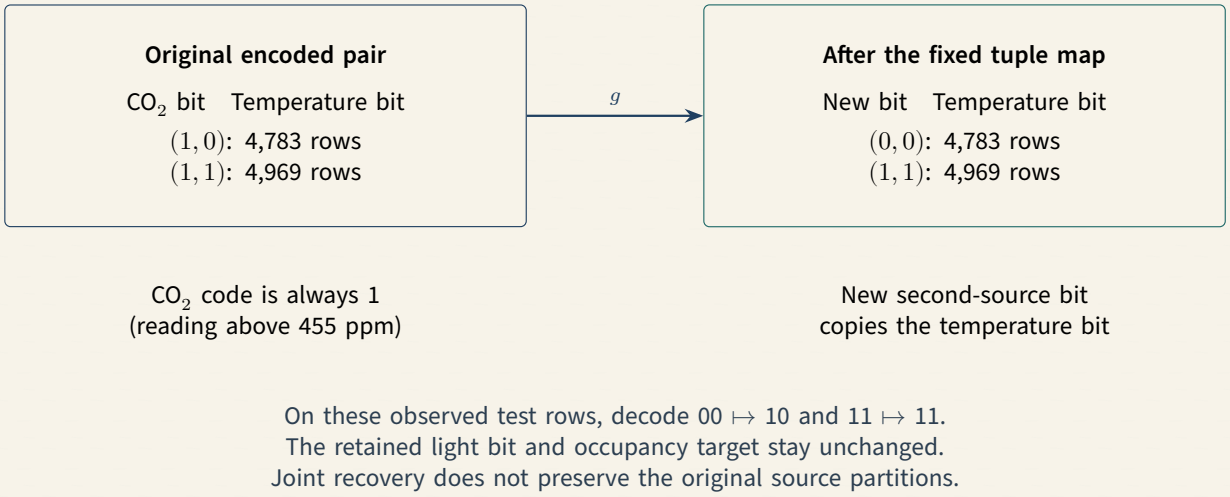


Figure 3: The observed office test recoding. The light bit and the target stay unchanged; recovering the pair does not preserve the partitions of the individual sources.

In (10), $J = \{3\}$ changes to $J' = \{2, 3\}$.

Quantity	Old node	New node	Signed value, nats
R_2	$\{\{1\},\{3\}\}$	$\{\{1\},\{2\},\{3\}\}$	0.178414142
U_X	$\{\{1\}\}$	same	0.069388796
U_Y	$\{\{3\}\}$	$\{\{2\},\{3\}\}$	−0.032706479
Q	$\{\{1,3\}\}$	$\{\{1,2\},\{1,3\}\}$	0.080019167

All other signed atoms are zero. With the unrounded values, (10) gives $D \approx 0.582279577$. The negative atom stays; it is not a causal importance of a sensor.

On the test days the fixed container uses 3,949 bytes, the lossless one 2,721 and the lossy one 2,718. Lossless coding costs only three header bytes more than lossy coding, at the same payload width. After decoding the original y, a receiver can still evaluate the same $G(x, g(y))$. It obtains the same predicted probabilities and keeps the original coordinates for attribution and audit. This is a derived software alternative, not another experiment, and it depends on the stated offline codebook, not on unknown future support. Choosing a model need not require discarding the original representation.

8. Evidence, failures and research process

AI role and review. The derivations were written and checked in AI-agent sessions. Several AI model critiques, each written before its author saw the others, reviewed the written proofs. The primary agent session then checked the premises and the files instead of counting agreement. No named human has checked the proofs. The proof by nested unions and the proof by refinement of one branch agree. The approaches compared included no action, mutual and conditional mutual information, direct prediction loss, lossless coding and evaluation on the complete lattice. Named review criteria describe what the review covered; they are not independent people.

On 28 October 2026 a further AI agent reviewed this report, its scripts and its data. A second session checked every claim of that review against the repository files or a new computation, and recorded the result in the adjudication record. The review was sound in its mathematics and in the defects it reproduced: two replay scripts failed, and the status statement of Section 10 was out of date. Its statistical discussion missed the points that Section 7 now explains. The two scripts were repaired, and the repaired runs reproduce all 121 published campaign files and the saved upper-bound check byte for byte; the addendum gives the details. This review and its adjudication are model output; their agreement is correlated, not independent evidence.

Formal checks. Z3 5.3.1 returned 46 unsat and 3 sat results for explicit event congruences, inequalities of source sets and signs, and controls that drop a premise. The 28 sign cases reduce to 14 coefficient forms. Exact matrices give the weights 22 and 67. These are checks of the algebra, not a formal probability theorem. The Z3 4.16.0 checks and the Lean versions that the repository fixes elsewhere did not change.

Exposure and selection. The data and their earlier results were known before the study. Algorithmic revisions kept the failed attempts and did not change the scientific predicates; no best of several scientific results was selected. The model family, the defining paper, the interpreter, the host, the owner and the file records are shared points of failure. The training data of the models are unknown, so agreement cannot establish originality or that no publication exists. Bounded searches found related work on invariance and partitions, not a priority result.

Rejected approaches. Kept as failures: equality of marginals or recovery of the whole vector as evidence that the atoms are preserved; costs in one direction only; pointwise extensions; changed targets or couplings with the wrong marginals; and bounds that ignore the number of values. The mean event surprise is not concave in general: probabilities $1/2$ on 00 and 11 give $G = \ln 2$, all probability on 01 gives zero, but their equal mixture gives $\frac{1}{2}\ln(4/3) < \frac{1}{2}\ln 2$. A draft in another source format and two attempts to export the figures also failed, and they are kept. No correction changed a scientific kernel, default, precision or expected outcome.

Open questions. A formal probability proof, the correspondence between the source paper, this report and the Rust code, sampling calibration and the five programmes A to E of a complete three-source certificate remain open.

9. Rust implementation and computational cost

The default `stable::categorical::discrete_sxpid_n` supports two to four sources. The new bounds and the exact comparison have no production Rust interface. The exact Python reference programs use three binary sources, up to four target values, $N \le 32{,}768$ rows and $Q_{\mathrm{occ}} \le 32$ occupied outcomes.

The medians cost $O(N \log N)$; counting and scoring cost $O(N)$. Direct source events cost $O(18Q_{\mathrm{occ}}^2 \cdot 3)$. Dense inversion needs quadratic storage and cubic arithmetic. Integer powers, greatest common divisors and series have costs that depend on the number of bits, with a limit of 1,048,576 bits per integer. A failed preflight, an expired deadline or an unresolved interval stops the run; the accuracy is never relaxed. The 48 dual certificates of bound (13) need 48×18 exact sums and a comparison per distribution. The exploratory polytope program solves a mixed-integer linear program per distribution in binary64.

The variations of information and the seven changes of mutual information can bracket the total change of the atoms before the atoms are computed. Uses with neural, imaging or embodied data need their own categorical mapping, paired transformation, target and validation. No neural inference, resampling, online control, sensor removal or energy experiment took place. No timing of a Rust implementation is reported.

10. Reproduction and citation

summary.csv covers all 30 distributions. The exact records keep the counts, products, intervals, class losses, matrices and the failed and successful attempts. This report was first published on 24 October 2026 and revised on 31 October 2026; both versions are on the main branch of the repository. Being on the main branch does not settle the open questions of Section 8.

Replay. Put the four raw recordings in the location that the scripts guide names. From the repository root, the following commands replay the study, replay the saved upper-bound check and check the dual certificates of bound (13):

```
V=audit/research/source-coupling-compression/verification
E=audit/research/source-coupling-compression/evidence
python3 -I -B $V/run_study_v3.py NEW_DIRECTORY
python3 -I -B $V/post_study_bound.py NEW_DIRECTORY_2
python3 -I -B $V/chain_lower_certificate.py
diff -r -x EXECUTION.json NEW_DIRECTORY $E/campaign
cmp NEW_DIRECTORY_2/REPORT.json $E/coupling-bound.json
```

The option `--raw-dir DIRECTORY` after the first argument of the first program selects another directory of raw data. The last two commands must report no difference: of the published campaign files, only the timing record `EXECUTION.json` differs between runs.

Method: Makkeh, Gutknecht and Wibral (2021), *Introducing a differentiable measure of pointwise shared information*, v5, version of record. Partitions: Meilă (2005), linked in Section 2. Re-encoding context: Matthias, Makkeh, Wibral and Gutknecht (2026), *Novel Inconsistency Results for Partial Information Decomposition*, Property 2. The data sources and their DOIs are in Section 6. Software: CITATION.cff.

Suggested citation: Sepehr Mahmoudian, *Source partitions and shared-exclusions information* (2026), with the exact containing commit or release. Cite MGW separately.